Delphi 39 way pinout for Speeduino adaptor board

Speeduino Pin#	Function	Delphi pin
1	Injector 1 - Pin 1/2	B1
2	Injector 2 - Pin 1/2	A1
3	Injector 3 - Pin 1/2	C2
4	Injector 3 - Pin 2/2	A2
5	Injector 4 - Pin 1/2	B2
6	Injector 4 - Pin 2/2	SPARE (INJ4)
7	Ignition 1	A4
8	Ignition 4	B4
9	Ground	GROUND (C1)
10	Ground	GROUND (C1)
11	MAP Sensor (0v-5v)	A5
12	Ground	GROUND (C1)
13	5v	B5
14	Proto Area 1 (0.4.4b+ Flex Sensor)	C3
15	Proto Area 2 (0.4.4b+ Fan)	C4
16	Proto Area 3 (0.4.4b+ Fuel Pump)	C5
17	Proto Area 4 (0.4.4b+ Tachometer)	C6
18	Proto Area 5 (0.4.4b+ Clutch)	C7
19	Coolant (CLT)	C8
20	Inlet Air Temp (IAT)	C9
21	O2 Sensor	A6
22	TPS input	B6
23	Ground	A11
24	Cam Input / VR2+	C12
25	Crank Input / VR1+	A12
26	VR2- (Not used for hall sensor)	B12
27	VR1- (Not used for hall sensor)	A13
28	5v	SPARE (5V)
29	Idle Stepper 2B	C10
30	Idle Stepper 2A	B11
31	Idle Stepper 1A	C11
32	Idle Stepper 1B	B10
33	Ignition 3	A7
34	Ignition 2	B7
35	Boost	A8
36	Idle 2 (For use with 3 wire idle valves)	B8
37	PWM Idle	A9
38	VVT	B9
39	Injector 2 - Pin 2/2	SPARE (INJ2)
40	Injector 1 - Pin 2/2	SPARE (INJ1)
X	12V Input via fuse	C13, B13

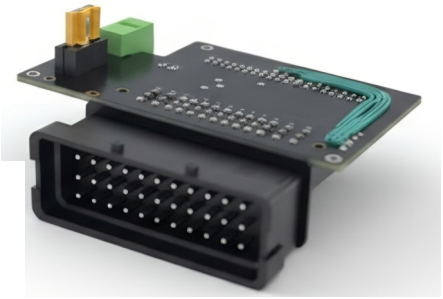
Areas to note;

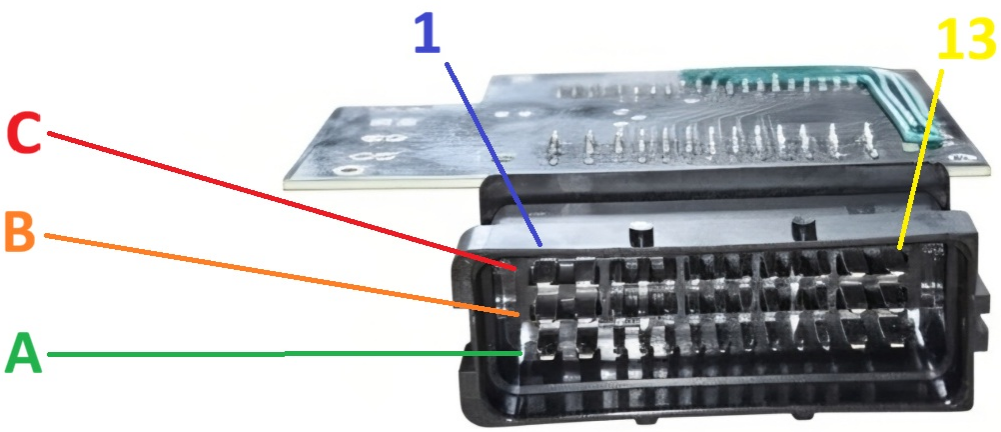
- Before committing to plugging your finished loom into the speeduino, always check with a multimeter where you expect to see voltage or ground.

- Unless specified, I have left the fuse and green power connector loose from the board, if you are using the case (STL) provided, the fuse mounts on the underside of the PCB as pictured below. These require basic solder skills.

- The green power connector will interfere with any daughter board placed in the stepper driver sockets. In this case, please solder your wires directly to the PCB and the other end should be connected to the 12v+ and ground main power supply at the back of the speeduino board.

- You have available to use, up to 7 spare free pins on the Delphi plug. These are **A3, B3, A10, B10, B11, C10** and **C11**. These can be used to solder jumpers from the optional areas of the adaptor PCB (secondary injector outputs, stepper controls for IAC , 5v/gnd or prototype opportunities).





Most questions can be answered from referencing this document alongside the Speeduino Wiki.

Otherwise please get in contact for any questions.

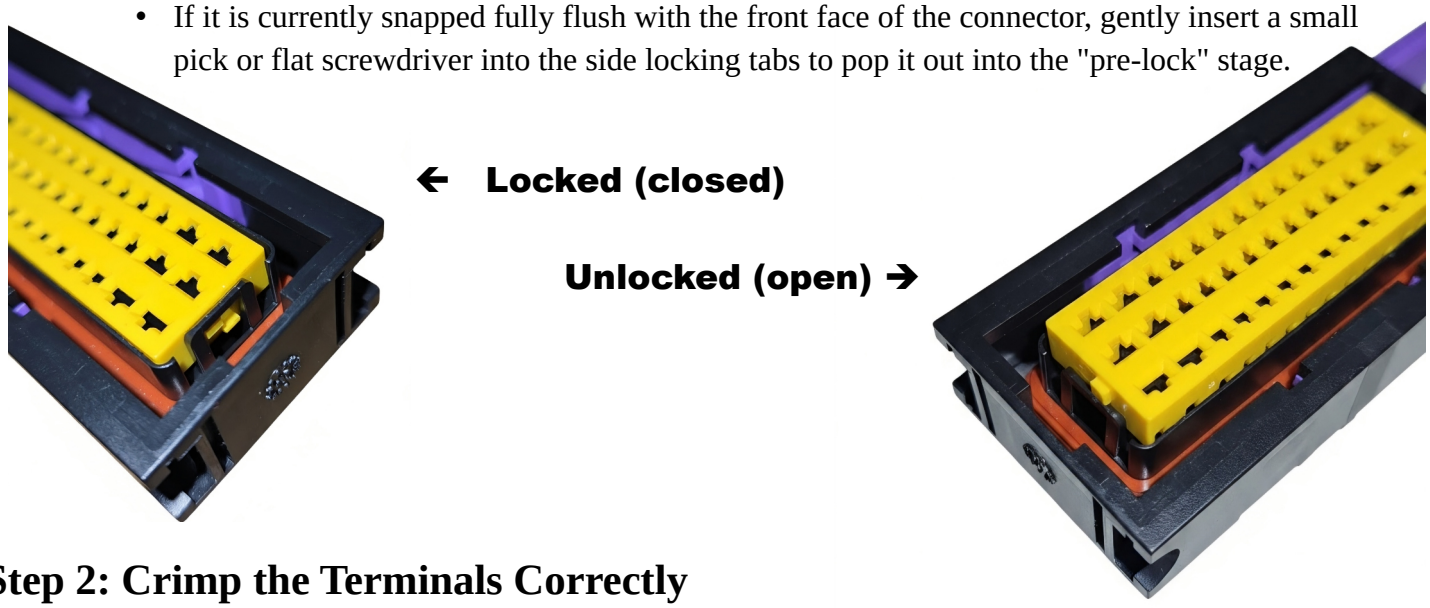
Step-by-step guide to properly assembly and pin a Delphi 39-way socket connector

Tools & Materials Needed

- **Delphi 39-Way Connector Kit (Included)**
- **Delphi/Aptiv Terminals (Included)** (typically Metri-Pack 150 series or Micro-Pack terminals).
- **Proper Ratcheting Crimp Tool** with the correct die set for open-barrel terminals.
- **Automotive-grade wire** (stripped 3–4 mm / 1/8 inch).
- *(Optional)* Small flathead screwdriver or terminal release pick.

Step 1: Prepare the Connector (TPA & Covers)

1. **Remove the Rear Wire Cover / Strain Relief:**
2. **Set the Yellow TPA to the "Pre-Lock" (Open) Position:**
    - **Do not fully remove the yellow face plate** unless necessary for cleaning or inspection.
    - The yellow face acts as the **Terminal Position Assurance (TPA)**. It needs to sit pulled forward/outward by about **1–2 mm** in its initial staging position.
    - If it is currently snapped fully flush with the front face of the connector, gently insert a small pick or flat screwdriver into the side locking tabs to pop it out into the "pre-lock" stage.



Step 2: Crimp the Terminals Correctly

1. **Strip the Wire:** Strip approximately **3–4 mm** of insulation off your wire. Avoid nicking or fraying any copper strands.
2. **Orient the Crimp:**
    - Insert the terminal into the correct die notch on your ratcheting crimp tool.
    - Place the stripped wire into the terminal so that the **front crimp wings** grab the bare copper wire and the **rear crimp wings** clamp tightly onto the wire insulation.
3. **Execute the Crimp:** Squeeze the crimp tool until it releases automatically.
4. **Inspect:**
    - Ensure both wings curl inward neatly into a "B" shape over the bare wire.
    - Verify the terminal body is straight and not bent or twisted.

Step 3: Insert the Terminals into the Housing

1. **Check Pin Orientation:**

   Examine the terminal end and match it to the rectangular slot geometry in the rear of the connector housing. The locking tab/catch on the terminal must align with the internal latching finger in the housing.

2. **Insert the Pin:**

   Push the terminal straight into the desired rear cavity until you hear or feel a distinct **click**.

3. **The "Tug Test":**

   Gently pull back on the wire to ensure the primary internal plastic tab has caught and locked the terminal in place. If the wire pulls out easily, check pin orientation or make sure the yellow TPA isn't accidentally pushed in.

Step 4: Lock the TPA (Secondary Lock)

1. **Verify All Pins:** Make sure every required pin is fully seated at equal depth.
2. **Engage the Yellow TPA:**

   Press the yellow face plate evenly from the front using your thumbs until it snaps **fully flush** with the connector body.

3. **Troubleshooting stiff TPAs:**
   - **If the yellow TPA refuses to push in flush easily: DO NOT FORCE IT.**
   - A stubborn TPA means one or more terminals are not seated fully forward. Pull the TPA back to "pre-lock", double-check each wire with a light push/pull, and re-seat any unlatched terminals.

Step 5: Final Reassembly

1. **Route Wires:** Align wires neatly through the rear cable exit.
2. **Reattach Cover:** Clip the rear plastic strain relief cover back onto the main connector housing.
3. **Check Lever Lock:** Ensure the lever arm moves freely through its arc before mating the plug to the socket or ECU header.

